Note → super() doesn't need calling self.
→ class. --init-- (self) needs self as the argument.

MRO → Method Resolution Order

class A (B, C):

⇒ MRO ⇒ A → B → C → object

If i call super() in A, it will go to B.__init__(). but not to C.

⇒ to use, C.__init__(self) is necessary.

Similarly, if 1 method is defined in B & C both, A's object will call B' defination by MRO

Magic methods (to define behaviour of objects, aka dunder methods)

eg: __init__() → initialises new instance
__str__: → returns a string representation
__repr__: → " official string "
__len__? → returns length of an object
__getitem__: → gets an item from a container
__setitem__: → sets " " " in a container
__add__(self, other): → regular addition
__sub__(self, other): → regular subtraction
__mul__(self, other): → multiply (*)
__truediv__(self, other): → divide (/)
__eq__(self, other): → == ?

__lt__(self, other): → <
__gt__(self, other): → >

2 arguments needed

Overwriting magic functions

~~def __str__(self):~~

consider a class Person:

```
def __str__(self):
    print('Hello')
```

person = Person()

print(person) → generally <object ... >

→ after overriding → 'Hello'

similarly, if we override __repr__(self):

→ call with print(repr(person))

---

class Vector:

```
    def __init__(self, x, y):
        self.x = x ; self.y = y.

    def __add__(self, other):
        return Vector(self.x + other.x , self.y + other.y)

    def __sub__(self, other):
        return Vector(self.x - other.x, self.y - other.y)

    def __eq__(self, other):
        return self.x == other.x and
               self.y == other.y
```

Vo Vo Imp.

can be used to manipulate vectors (x, y)

eg: V1 = Vector(2,3) ; V2 = Vector(4,5)

print(V1 + V2) , (V1 - V2) or V1 == V2

→ (6, 8) , (-2, -2) , False

Custom Exceptions

```
class Error (Exception):
    pass

class dobExcept. (Error):
    pass

# In any part of code, where there might be
# such an error, eg.

try:
    if age <= 30 and age >= 20:
        pass
    else:
        raise dobExcep.

except dobExcep:
    print ("Sorry, age not qualify")
```

Property decorators → study, not in notes.

- @ property
- @ getter
- @ static method.
- @ x. setter

??